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Fair-Compass SFV

A Proposed Intelligent Interactive System for Reducing Content Bias in Short Form Social Platforms

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Abstract

Many popular Short-Form Video (SFV) platforms such as TikTok, Instagram, and YouTube are designed to be engagement-first, using recommendation systems that do not account for content fairness, leading to feedback loops and, over time, echo chambers. Fair-Compass SFV (FC-SFV) addresses this by implementing a fairness-aware recommender system trained exclusively on non-recommendation-driven interaction data from an existing SFV platform, reducing popularity bias. Using lstmDeepFM with a fairness constraint layer, FC-SFV models both long-term and short-term user preferences, adapting recommendations to genuine individual engagement rather than group popularity.

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1 Problem Statement

Short-Form Video (SFV) platforms are consumed by adults averaging 51 minutes per day [1], yet the recommendation systems that act as the backbone for these are under-performing for user wellbeing and content diversity, with high-intensity videos that maximise for engagement being vastly favoured over educational and low-intensity content [2].

Algorithms are the main driver of the type of content that is consumed. In the systematic literature review covering bias and fairness [3], it was found that there were a few different areas responsible for affecting content recommendation biases including: lack of reliable metrics; dark patterns that discourage natural stopping points; and echo chambers caused by repeated feedback loops. The effect of echo chambers is significant as it can cause belief reinforcement, whereby users are exposed exclusively to content that aligns with their existing beliefs, strengthening viewpoints, and increasing isolation from other communities [4]. Additional effects include misinformation vulnerability, caused by reduced analytic thinking, users in echo chambers are less likely to question content when the same claims are repeated across multiple sources [7]

Algorithms common in SFV platforms have been shown to reduce users' analytic thinking and overall are less likely to detect misinformation [7] — This is particularly an issue with younger audiences (aged 18-34) who use social media as their primary source of news [1]. The psychological impact cannot be fully self-corrected by user intervention such as scrolling away or negatively reacting.

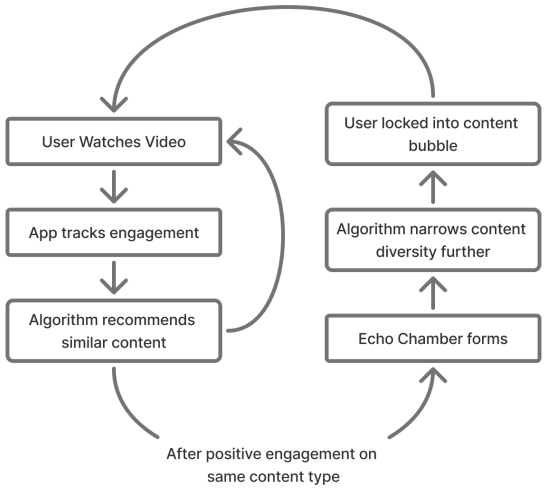


Figure 1: Feedback loop formed by unoptimised engagement-first algorithms

Existing solutions attempt to resolve excessive consumption of SFV [5,6], but such interventions treat the symptom rather than the inherent cause, leaving the underlying algorithmic biases in content diversity up to the social media platform [3]. One such example adds an intention screen before starting a session and an ability to reflect on every video that you watch [5]. Such a solution, would allow for a more suitable place to stop watching but breaks viewing experiences through harsh pauses that act more as annoyances than as places of learning. Moreover, these types of solutions are reliant on the content being shown to be accurate and educational otherwise it may strengthen the impact of misinformation.

Rather than removing engagement signals to have a completely fair system, an improvement would be to optimize engagement metrics to recommend content that the user has a positive interaction with instead.

2 Proposed Solution

2.1 System Architecture

Fair-Compass SFV (FC-SFV) is a fairness-aware short-form video recommender system designed to correct for popularity bias in training data and reduce the effect of feedback loops.

2.1.1 Dataset

KuaiRand-Pure has been selected as the dataset for FC-SFV due to its data collection methodology, using a random-exposure mechanism, allowing for unbiased training data for the system [8]. The pure variant of the dataset has been selected from the full dataset KuaiRand-27K, the dataset collected from the Chinese SFV platform Kuaishou (https://www.kuaishou.com/). The recommender system used in Kuaishou works by queuing videos based on collected user behaviour data. For 0.37% of served videos, a randomly selected video is shown to the user irrespective of the predicted user preference — In the pure dataset, this non-recommender based content served to the user is extracted, giving a final set size of 1,436,609 interactions across 27,285 users [8].

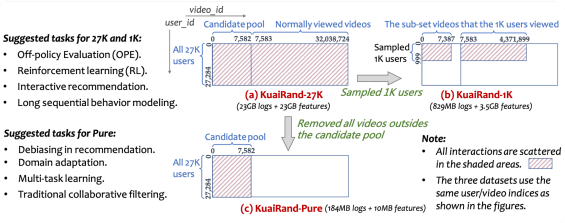


Figure 2: Visualisation demonstrating how data is split across the three available KuaiRand datasets (27K, 1K, Pure)

For Short-Form Video recommenders, it is important to test on a dataset that is both unbiased and from the SFV domain in order to get relevant metrics. Unlike long-form video, the metrics to measure engagement and likeability are much more limited and can be easily swayed by superficial signals such as opening the comments or watching the video briefly [10] — KuaiRand-Pure uses 12 signals (watch time, hated video, followed, time spent in comments, etc.) to improve granularity and reliability of determining positive engagement.

When tested against other datasets that are unbiased, there is no other in short-form videos especially at this scale. Comparing it to alternatives, KuaiRand beat out other unbiased datasets for recommenders: Yahoo!R3/R6A/R6B; Coat; and Open Bandit — It has more feedback signals, features, timestamps and has a larger set of items and interactions [8].

2.1.2 Model

DeepFM uses a Factorisation Machine (FM) component and a deep neural network together combined using a shared-embedding architecture, allowing it to capture both low-order feature interactions such as user preferences for a specific category of content, and high order interactions such as the relationship between different demographics, location, and history [11]. Unlike with FNN which requires pre-training and Wide & Deep which requires manual feature engineering, DeepFM works without either, using a shared embedding layer instead — for an unbiased system, it is important that no human interference can interfere with the recommender.

lstmDeepFM with LSAB has been selected as the model for FC-SFV, combining DeepFM with Long Short-Term Memory (LSTM) for the purpose to optimize click-through-rate (CTR) [12]. Learning Self-Attention Bias (LSAB) is added to the self-attention MLP, acting as a bias strategy [13]. Unlike long-form video (LFV) where the user is able to filter through recommendations to find a video to watch, SFV continuously serves videos to the user — base Deep Factorization Machines (DeepFM) treats each interaction as independent, ignoring this dependency entirely. By applying LSAB to the self-attention mechanism within the Self-Attn MLP component of lstmDeepFM, it is able to model user preferences over time, allowing sudden changes in viewing preferences to have more impact in recommendations.

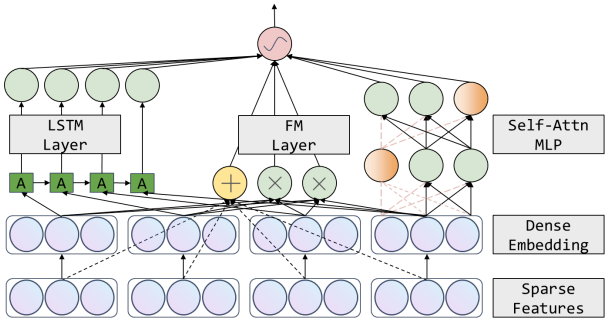


Figure 3: Improved DeepFM-Based Model with LSTM.

2.1.3 Fairness Framework

SaFair is a model-agnostic fairness framework [9], making it a compatible choice to work with lstmDeepFM as the model and thus has been selected to act as a further fairness layer within FC-SFV for the purpose of penalising the model whenever it over or underestimates a group's preferences compared to the actual posterior ratings. When SaFair was added to other base models such as DNN, DCN and DeepFM, it consistently outperformed the base by between 0.2% and 0.7% when tested on KuaiRand-Pure [9]. Although a minor improvement, over a large scale, this performance improvement can significantly improve recommendations for the user — it should be noted that although SaFair was tested for DeepFM, since the model architecture remains the same, the fairness gains are expected to carry over to lstmDeepFM.

SaFair estimates how closely a user's taste aligns with their group's preferences, scaling the group constraint based on the user, allowing a user who has a different behaviour compared to the group they have been determined to be in to get a softer constraint applied. In practice, this allows recommendations to reflect genuine individual engagement over defaulting to group popularity. Balancing group and individual signals prevent atypical users from being recommended content that doesn't reflect what they want, helping to counteract feedback loop formations, with deviations in content viewing experiences adjusting the algorithm's weighting.

2.2 Algorithmic Pipeline

2.2.1 Data Pre-Processing

From the dataset KuaiRand-Pure, the 12 signals are preprocessed into a unified engagement score for each interaction, representing if the user found the content to be positive or negative based on their actions. A weighted combination will be applied based on a variety of different action groups that are performed (e.g. follow creator and check comments would be a highly positive interaction). By analysing the action groups performed over the final metrics, further context can be added into the reaction out of the video [10]. The final normalised scalar engagement score can be used as an additional metric to determine whether to aid in the training process.

To protect against popularity bias, features such as like counts and number of shares would be excluded from the item features training dataset. By excluding these, it ensures that the model is able to learn from user-item engagement rather than popularity-driven exposure patterns [14].

Item and user metadata is additionally included as encoded input features, passed into the shared embedding layer of lstmDeepFM alongside the engagement score . Additionally 18 encrypted categorical user features are included, labelled onehot_feat0 to onehot_feat17, which may include attributes such as device type, age bracket, and location [10]. These are treated as learnable embeddings and can help the model identify correlations between users to aid in the formation of user groups.

2.2.2 Model Training

The pre-processed data is passed to lstmDeepFM where the user's interaction history is processed to capture how their preferences shift over time in order to learn behavioural changes. By using LSTM, long-term and short-term patterns are learnt to ensure accurate viewing intent — the final result is a model that produces a click-through rate (CTR) prediction score for each user-item pair [11]. During

the training process, SaFair adds a fairness penalty to the loss whenever the model over or underestimates a group's preferences, scaled per user so outliers are not overcorrected for.

2.2.3 Personalised Recommendations

When a user begins their session, a series of potential videos to show the user is generated — to get to that, the recommender system works through a two step process. In the first, a two-tower retrieval model narrows the full video archive into a small candidate pool by encoding the user's current session state and each video's features into separate embedding spaces, using dot product similarity to determine the candidates [15]. In the second step, the trained lstmDeepFM model scores each of the candidates. To address the issue of sudden preference shifts, a lightweight model is triggered every time that the user swipes to the next video, taking the real-time feedback signals within the current session to reorder the remaining candidates accordingly [16]. Within this reranking, SaFair adjusts the ranking accordingly to ensure content fairness.

For newly uploaded videos with little to no prior interaction history, content features such as category tags and video transcript are used instead. Without this step, new content risks being excluded from the content pools or being over-recommended if the little amount of data is positive [14].

2.3 User Interface

FC-SFV is designed with a transparency-first set of controls, allowing the user to control how the algorithm learns what they want. A set of buttons on the main screen include: like, follow, profile, share, comments, "not interested" and bookmark — these are standard to many of the leading SFV platforms. Passive interaction signals such as watch percentage & time, rewatch count, profile visits, discovery source, time spent in comments, combined with video metadata are used to create an engagement score described previously and place the user within a group.

Where FC-SFV differs from other popular platforms is the inclusion of a diversity slider within the settings, allowing users to adjust how strongly their recommendations are influenced by their assigned user group versus their own individual behaviour. On the backend, this directly scales the SaFair fairness constraint — a lower setting applies a softer group constraint, prioritising personal viewing history, while a higher setting aligns recommendations more closely with the broader group, naturally introducing content the user may not have normally encountered.

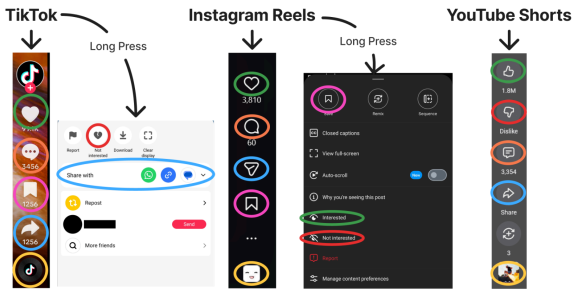


Figure 4: Comparison in current interaction buttons within TikTok, Instagram Reels and YouTube Shorts. Green: Positive; Red: Negative; Orange: Comment; Blue: Share; Pink: Save; Yellow: Audio.

3 Critical Analysis

3.1 Algorithmic Limitations

One of the main limitations of FC-SFV is the cold-start problem. Despite implementing fallbacks for new videos, when there is little data to base the recommendations off of, the algorithm can struggle to have enough multi-group analytics to determine if it is a good recommendation [14]. Such examples include videos without speaking and when the creator has not put in any metadata themself — one potential mitigation, though computationally expensive, is to infer thematic context from the creator's existing content or applying automatic subject detection [8]. Neither approach has been implemented in FC-SFV as both are impractical for the reasons above but instead, a similar approach used in Kuaishou can be used, showing the user a

new video to a small subset of each group in order to get data for the wider user base.

Additional limitations include the problem with using an LSTM-based model of DeepFM — users with limited histories with the platform may get reduced personalized videos served and be shown popular videos as these would have the highest multi-group engagement. This is caused by LSAB being unable to model behavioural patterns based on limited data [13]. A limitation such as this is practically impossible to completely fix and thus FC-SFV is not further optimised to prevent.

The engagement score used in training is defined on manual weighting of the twelve KuaiRand feedback signals, meaning that for an interaction, for a specific action group, they may not be accurate to whether the user in fact actually had a positive or negative engagement with it. Different metric weighting could produce vastly different ranking outcomes. To reduce the effect of this, multiple A/B user testing phases can occur during the development to find the optimal weighting.

SaFair's fairness constraints are dependent on creator group labels being available and well-defined [9] — In the case of it being incomplete, fairness may not be guaranteed. This is further exemplified as new creator demographics or content categories emerge that did not exist in the training data. Without retraining the model to keep these into account, SaFair will be unable to be fair.

3.2 Ethical and Societal Considerations

The dataset used (KuaiRand) comes from Kuaishou, a Chinese social media platform [8]. By having this be the dataset, it adds bias into it by the sheer nature that the majority of the users who use it are Chinese, people who may have different preferences for what content they may watch or the way that they interact with SFV platforms. Additionally, due to using a specific social media platform as the basis, the feedback signals used to train it is specific to that platform — implementing that in another platform which may have different buttons and ways to interact may cause conflicts and may struggle to interpret what the correlation of the signals it has trained on versus the one of the new platform.

The current implementation of FC-SFV does not take into account the filtering out of harmful or hateful content, only to show content with positive engagement and reduce content bias. Without these in place people who are looking for that harmful content may be able to find it more easily than on other platforms both because it doesn't have a filter but additionally because it is better at showing content that the user is interested in. On top of this, new videos are more likely to show this content to people who are not interested in it due to it looking for video content rather than engagement and as a result has the potential to be an ethical concern regarding the Online Safety Act. A fix would be to filter content when it is being uploaded to prevent this sort of content from being recommended at all.

To comply with GDPR, FC-SFV will have profiles capable of being erased (Article 17). When someone requests to be forgotten, it will delete their account with their interaction history and any other data collected. Due to the black-box nature of lstmDeepFM, it is not possible to fully explain why specific content was recommended (Article 22). Instead, the user will be able to see which group they fell under and an explanation will be provided explaining how the recommendation system works. Sensitive data will be anonymised locally and sent to be used to create personalised results as described in 2.2.3 .

4 Conclusions

Fair-Compass SFV (FC-SFV) aims to tackle the problem of content bias in short form video by implementing a multi-layer pipeline where it is taken into account at each step. The dataset, KuaiRand-Pure, has had no algorithmic recommendation behind it, allowing for true random user interactions. lstmDeepFM is the main model behind it all which uses long-term short-term memory combined with DeepFM to reduce popularity bias in the content served and instead show content that is better aimed at the user, taking

into account both the long-term and short-term patterns. LSAB is used to allow for sudden changes in user preferences, prioritising the short-term over long-term. SaFair is a fairness layer that further protects against overfitting grouping by penalising overconfidence.

FC-SFV fails to adapt to current trends and grouping, using static weighting which makes it fail over time without retraining the model. Additional work could expand the dataset and have the model relearn user patterns itself as more users use it, learning the overall trends and patterns.

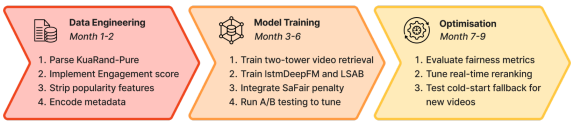


Figure 5: Approximate timeline for implementation of FC-SFV in real-world scenario

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